

When AI Agents Talk, What Information Gets Lost?

A controlled study of independent analysis, debate, and source-grounded criticism

Using CA-TradeoffEval with a cross-model replication

Spriha Pradeep

Independent Researcher | ORCID iD: 0009-0009-9451-7660

September 2026

Preprint (not peer reviewed) | DOI: 10.5281/zenodo.22715133

Main finding: In the 10-measure confirmatory study, adding a source-grounded critic improved information preservation over ordinary debate by 3.13 percentage points. A five-measure Claude replication produced a similar +2.62-point estimate, although with wider uncertainty. Ordinary debate did not reliably outperform simply combining independent analyses in either model family.

Abstract

When many AI systems work on the same problem, it is intuitive to think that letting them discuss their answers will naturally make the final result better. But this discussion can also cause useful details to disappear as the systems converge on one shared response. I tested this possibility in long form policy analysis, where it is necessary for a good answer to keep track of many facts and trade-offs. I built CA-TradeoffEval, an evaluation testbed based on official California ballot-measure materials, and broke ten policy measures into 120 small reference claims and 367 smaller information components. I compared four ways of producing a final answer: one AI agent working alone, three independent agents whose work was combined, ordinary multi-agent debate, and debate in which one participant acted as a critic whose job was to catch missing or weakened information by referring back to the source. Across 160 final outputs from GPT-5.6 Sol, critic-guided debate preserved more source-grounded information than ordinary debate by 3.13 percentage points (95% measure-level bootstrap CI [0.53, 6.34]). Independent aggregation had a smaller positive estimate over a single agent, and ordinary debate was almost indistinguishable from independent aggregation. I then ran a smaller replication using Claude Sonnet 5 on five randomly selected measures. Claude had a similar critic effect (+2.62 points), but the confidence interval was wider and included zero. The Claude replication also showed more distortion in critic debate, suggesting that there is a possible trade-off between keeping information but also possibly introducing more inaccuracies. Overall, the results suggest that how AI agents are organized matters, making them discuss is not enough, and a clearly defined critic role can help preserve information, but it should also be paired with checks for distortion.

1. Introduction

1.1 The basic problem

Imagine asking three people to read and analyze the same complicated policy document. One person might notice the funding details, another might notice the timeline, and a third person might notice a small exception buried somewhere near the end. If they write separately, those details all exist somewhere. If they then discuss the issue and produce one joint answer, some details may survive, some may be combined, and some may just quietly disappear. My project asks if the same thing happens when multiple AI agents work together.

Large language models (LLMs) are being used increasingly not only as single chatbots, but as parts of multi-agent systems. In a multi-agent system, separate instances of a model are assigned roles, generate their own analyses, read one another's work, revise, or help produce a shared final answer. Researchers have proposed debate and other multi-agent methods as ways to improve reasoning, factuality, and oversight (Irving, Christiano, & Amodei, 2018; Du et al., 2024). But most evaluations still focus on whether the final answer is correct. These evaluations also largely focus on objective questions with only one correct answer, not forms of open-ended analysis. That process can miss a different failure mode, which is that the final answer can still sound reasonable while dropping useful information that one of the agents originally noticed.

That matters in tasks like policy analysis because there is rarely one short 'correct answer.' A useful policy explanation may need to preserve fiscal effects, who is affected, implementation dates, legal limits, arguments for and against a proposal, and uncertainty about downstream consequences. Two answers can recommend the same thing but give a reader very different pictures of the trade-offs. In this setting, omission is very important.

I call the measurable version of this problem deliberative information loss: information that is relevant and supported by the source but is missing, materially weakened, or distorted in the final synthesized answer.

1.2 What I mean by an “AI agent”

In this paper, an AI agent is one model instance given a specific prompt and role. For example, three agents can all receive the same policy source and independently write analyses. A later agent can then be asked to synthesize those analyses. Another agent can be assigned the role of critic and told to look specifically for facts, qualifications, or trade-offs that the others missed. Therefore the important experimental variable is not the number of agents or the type of model, it is how the same underlying model is organized into a workflow.

1.3 Why compare independence, debate, and criticism?

Independent sampling is already known to be useful in some settings because different runs can make different mistakes and notice different information. Methods such as self-consistency can take advantage of this variation (Wang et al., 2023). Debate adds a new aspect, which is that agents are exposed to one another's reasoning before the final synthesis. That could help them correct mistakes, but it could also reduce diversity or encourage convergence. Research on the 'wisdom of crowds' has similarly shown that social influence can reduce the benefits of independent judgments in humans (Lorenz et al., 2011). The analogy is imperfect, but it still motivates the question: what changes when model outputs become socially connected rather than independent?

Recent work gives mixed reasons to expect both benefits and failures. Multi-agent debate has improved reasoning and factuality on several benchmarks (Du et al., 2024). Other evaluations find that debate is not consistently better than strong single-agent or sampling baselines, and some studies report conformity-like behavior or cases in which agents move away from correct answers after seeing peer reasoning (Zhang et al., 2025; Weng, Chen, & Wang, 2025; Wynn, Satija, & Hadfield, 2025). These results suggest that the details of the protocol, like who sees what, when, and with what responsibility, could actually be more important than the very broad label of 'multi-agent debate'.

My intervention design is deliberately simple. In the critic condition, one revision-stage agent is not told to participate normally. Instead, it is told to compare the developing analysis against the source and identify information that has been omitted, weakened, or stated without enough support. This keeps the number of agents and revision-stage outputs constant while changing the role structure.

1.4 Research question and predictions

Primary research question: When AI agents analyze the same policy source, how do independent aggregation, ordinary debate, and source-grounded criticism affect how much decision-relevant information survives in the final answer?

- Prediction 1 - Independent aggregation: combining several independent analyses will preserve more information than using one agent alone.
- Prediction 2 - Interaction effect: ordinary debate may preserve less information than independent aggregation because agents can converge and drop details during revision.
- Prediction 3 - Critic effect: replacing one ordinary revision with a critic will improve preservation relative to normal debate.
- Prediction 4 - Quality trade-off: keeping more information is not automatically better if the extra material is distorted, so preservation and distortion should be evaluated together.

1.5 What this project contributes

The project makes two contributions. First, it separates three things that are bundled together often, like getting multiple independent samples, allowing agents to interact, and assigning one agent a critical role. Second, it introduces CA-TradeoffEval, an evaluation testbed for measuring whether important policy information survives into a final answer. I call it an evaluation testbed rather than a validated benchmark because it currently covers one domain and was built primarily by one researcher. The point of the current study is to make one specific failure mode measurable and reproducible.

The longer-term theory of change is that if we can measure where multi-agent workflows lose or distort useful information, developers can compare architectures before using them in decision making settings. A useful system should not just reach a plausible conclusion; it should also be preserving enough of the relevant evidence and trade-offs for a person to understand why that conclusion is reasonable and what factors could also count against it.

2. Related Work

2.1 Multi-agent reasoning and debate

Multi-agent language-model systems use multiple model calls to solve a task through independent generation, communication, critique, or synthesis. Debate is one approach to that. Du et al. (2024), for example, showed that exchanging reasoning can improve accuracy and reasoning on some tasks. Earlier work on AI debate also proposed adversarial discussion as a possible way to make reasoning more visible to a human judge (Irving et al., 2018). Other collaborative designs, including cross model communication, explore similar ideas (Yin et al., 2023).

But better performance is not guaranteed. Recent work questions whether multi-agent debate consistently beats simpler baselines and shows that interaction can sometimes push agents toward one another’s conclusions and towards premature consensus without actually improving their correctness (Zhang et al., 2025; Weng et al., 2025; Wynn et al., 2025). This is why my study includes an independent-ensemble condition. Without that comparison, any gain from debate could just be a gain from having more model samples available.

2.2 Measuring long-form answers

Long-form answers are difficult to evaluate because one overall score can hide many smaller successes and failures. The FActScore system addresses this for factuality by breaking a generated answer into atomic facts and checking them individually against a source (Min et al., 2023). Similarly, HELM argues that language-model evaluation should use multiple clearly defined metrics instead of treating quality as one number (Liang et al., 2022). I borrow the same general idea of atomic evaluation, but apply it to preservation: rather than asking whether every statement the model makes is factual, I ask whether a fixed set of source-grounded information components is still present in the final answer.

This difference is important because a model could produce a perfectly factual but incomplete answer. It could also preserve many relevant details while introducing some distortions. My scoring therefore keeps preservation and distortion separate.

2.3 Why this is an evaluation testbed, not yet a general benchmark

Benchmarks are useful only when their scores actually represent the capability that people think they represent. Raji et al. (2021) warn that narrow datasets are often treated as broad measurements of intelligence or competence without enough evidence for that. CA-TradeoffEval is narrower. It tests information preservation in a bounded set of California policy documents under controlled multi-agent workflows. More future work here would need possible independent annotation, additional domains, and more model families before I can claim broad benchmark validity.

3. Methodology

3.1 Overview

The experiment used ten California statewide ballot measures as ten separate test cases. For each measure, I created four experimental conditions and ran four paired trials per condition, producing 16 final answers per measure and 160 final answers in Study 1. The same source packet and the same underlying information inventory were used across conditions within a measure. The final answers were then scored and condition identities were hidden.

3.2 Source material and CA-TradeoffEval

The source material came from official California Secretary of State voter-information materials. I chose official voter guides because they are bounded, public, and designed to describe complicated policy choices in a relatively standardized format. They also contain multiple types of decision-relevant information including fiscal effects, legal changes, implementation details, arguments, and qualifications, and this makes them useful for studying information coverage.

For each measure, I converted the source material into a reference inventory. A reference claim represents one meaningful piece of policy information. These claims were then divided into atomic components when a single claim contained multiple facts that could be preserved or lost separately. Across the ten confirmatory measures, the inventory contained 120 claims and 367 atomic components.

Measure	Atomic components	Study 1 scoring rows
M02	65	1,040
M03	58	928
M04	12	192
M05	64	1,024
M06	29	464
M07	22	352
M08	28	448
M09	25	400
M10	31	496
M11	33	528
Total	367	5,872

Table 1. Size of the confirmatory evaluation inventory.

3.3 What counts as preserved?

Each final answer was checked against each atomic component for its measure. Preservation was scored as 1 when the required information was correctly and sufficiently present, and 0 when it was absent, contradicted, materially wrong, or too vague to count. The goal was not exact wording. A model could paraphrase freely as long as the underlying information survived.

I scored distortion separately on a 0–2 scale: 0 for no meaningful distortion, 1 for a minor distortion, and 2 for a material distortion. Keeping the two concepts separate matters because an answer can mention a topic while stating it inaccurately.

The main preservation metric was claim-weighted preservation. Within each reference claim, I first calculated the fraction of its components that were preserved. I then averaged across claims so that a long claim with many components did not automatically count more than a shorter claim. As a robustness check, I also reported component-weighted preservation, which simply averages across all atomic components.

3.4 The four experimental conditions

Condition	What happened	What it tests
Single agent	One initial analysis was used as the final answer.	Baseline: what one model sample preserves.

Independent ensemble	Three agents analyzed the source independently. A synthesizer combined their work without a debate stage.	Whether multiple independent samples improve coverage.
Normal debate	The same three initial analyses were shown to agents, which produced ordinary revisions before synthesis.	What interaction adds beyond independent aggregation.
Critic debate	The setup matched normal debate, except one revision role became a source-grounded critic looking for missing or unsupported information.	Whether structured dissent changes preservation.

Table 2. Experimental conditions in plain language.

My design intentionally reused the same shared initial analyses across the relevant conditions to make the comparison between independent aggregation, normal debate, and critic debate more controlled. The conditions begin from the same discovered material, and then the later interaction structure changes. In the critic condition, two ordinary revisions were reused from the normal-debate setup and only one revision role was replaced by the critic. The final synthesis prompt was kept exactly the same.

3.5 Trials and pairing

Each measure had four paired trials. Pairing matters because different generations can vary a lot even under the same prompt. Rather than comparing unrelated random outputs, I compared conditions within the same measure and trial structure. The main statistical unit was the policy measure, not the thousands of component-level scoring rows. This avoids pretending that every component is an independent experiment.

3.6 Blinding and human verification

Scoring was conducted before I knew which blinded answer belonged to which experimental condition. Final outputs were assigned blind IDs, and the condition mappings were sealed. A model-assisted scoring system generated preliminary component labels, evidence quotes, and flags, but I manually reviewed every single final component decision. The preliminary scorer was used to speed up annotation and not to replace human verification.

The scoring pipeline also used exact-quote validation, for example, when the automated scorer claimed that a component was supported by a quote, the quoted text had to appear exactly in the blinded answer. After review, every measure was locked into an immutable scoring file. Only after all ten measures were locked and an integrity audit confirmed 5,872 complete rows with zero unresolved flags did I open the condition mappings.

This procedure was designed to reduce a simple but serious risk: if I knew which answers came from the critic condition while scoring them, my own expectations about the hypothesis could influence close judgments. Blinding does not necessarily eliminate every source of subjectivity, but it makes that pathway much harder.

3.7 Statistical analysis

For each measure and condition, I first averaged the four trials. Then I calculated three planned paired contrasts: independent ensemble minus single agent, normal debate minus independent ensemble, and critic debate minus normal debate. The primary outcome was claim-weighted preservation. Component-weighted preservation and distortion were the secondary outcomes.

Uncertainty was estimated with 10,000 paired bootstrap resamples over the ten measures using seed 20260809. In plain language, the bootstrap repeatedly resamples whole policy measures and asks how much the estimated difference changes. The resulting 95% interval describes the range of effects compatible with those resampled measure sets. I also ran leave-one-measure-out checks: each measure was removed once to see whether any single policy measure determined the direction of the result.

As for distortion, I reported three pre-specified summaries separately: average component distortion severity, the share of components with any distortion, and the share with material distortion. I did not combine them into one score.

3.8 Study 2: cross-model replication

Study 1 used GPT-5.6 Sol. After Study 1 was fully scored, locked, unblinded, and analyzed, I ran a separate replication using Claude Sonnet 5. Because I already knew the Study 1 results, I treat this as a post-confirmatory external replication rather than pretending it was part of the original confirmatory design.

Before generating Claude outputs, I froze a Study 2 protocol. Five of the ten measures were selected by a deterministic random procedure using seed 20260809: M03, M08, M02, M09, and M05. The replication kept the same four conditions, four paired trials per measure-condition, frozen component inventories, scoring guide, and scoring prompt. It produced 80 final Claude outputs and 3,840 human-reviewed scoring rows. Study 2 remained blinded until all five measures were locked.

4. Results

4.1 Study 1: overall preservation

The clearest Study 1 result was that critic debate preserved more source-grounded information than normal debate. Mean claim-weighted preservation was 0.8803 for the single-agent condition, 0.8929 for the independent ensemble, 0.8902 for normal debate, and 0.9214 for critic debate.

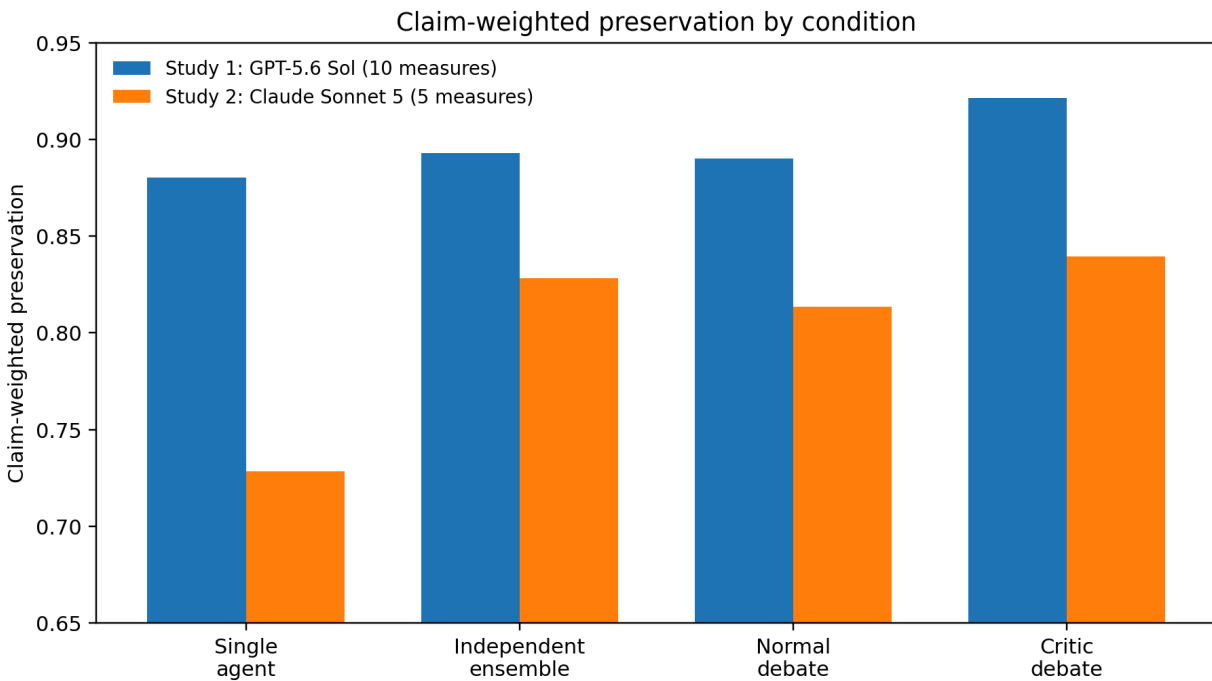


Figure 1. Claim-weighted preservation by condition in Study 1 (GPT-5.6 Sol) and Study 2 (Claude Sonnet 5).

The independent ensemble exceeded the single-agent baseline by 0.0126, or 1.26 percentage points. Its 95% bootstrap interval was $[-0.0002, 0.0260]$, so the estimate was positive but uncertain. Five of ten measures favored the ensemble, two favored the single agent, and three were tied.

Normal debate was 0.0027 lower than the independent ensemble. Its interval $[-0.0096, 0.0040]$ included zero. Four measures favored normal debate, five favored the independent ensemble, and one was tied. This does not support a strong claim that debate systematically destroys information, instead, it shows that ordinary debate did not reliably improve preservation beyond simply combining independent analyses in this experiment.

The critic condition produced the strongest difference. Critic debate exceeded normal debate by 0.0313, or 3.13 percentage points, with a 95% bootstrap interval of $[0.0053, 0.0634]$. Seven of ten measures favored critic debate, one favored normal debate, and two were tied. This was the only primary Study 1 contrast whose interval stayed fully over zero.

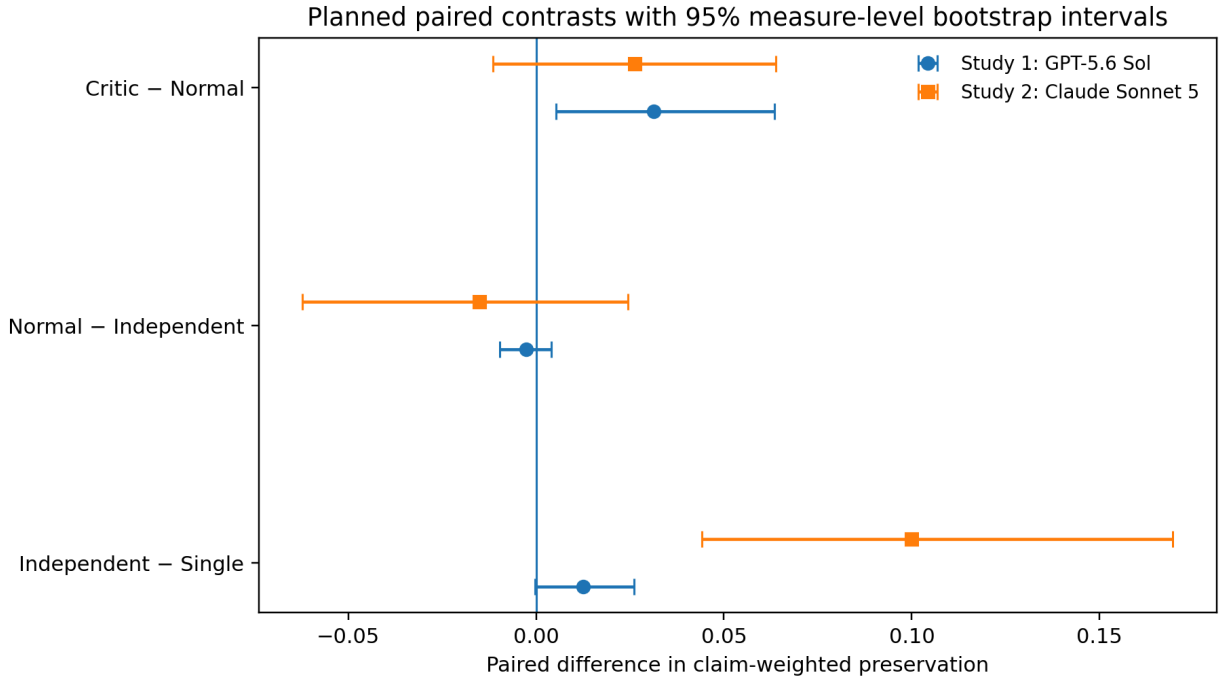


Figure 2. Planned paired preservation contrasts in Study 1 and the five-measure Study 2 replication.

4.2 Robustness checks

The critic effect was not an artifact of the claim-weighting scheme. Under component weighting, critic debate exceeded normal debate by 0.0352, with a 95% interval of [0.0097, 0.0678]. Leave-one-measure-out analysis also kept the critic-minus-normal estimate positive no matter which policy measure was removed, the estimate ranged from 0.0188 to 0.0363. The other two contrasts also kept the same direction in leave-one-measure-out analysis, although their bootstrap intervals still included zero.

4.3 Distortion in Study 1

Distortion was uncommon in Study 1. Mean component distortion severity ranged from 0.0148 in the single-agent condition to 0.0289 in critic debate. The planned bootstrap intervals for differences in mean severity and any-distortion rate included zero. Material-distortion differences were also very small. Study 1 therefore did not provide clear evidence that the preservation gain from the critic came with a systematic distortion cost.

4.4 Study 2: Claude replication

The Claude replication produced a broadly similar pattern, but with an important new qualification. Claim-weighted preservation was 0.7285 for a single agent, 0.8284 for the independent ensemble, 0.8134 for normal debate, and 0.8397 for critic debate.

The primary critic-minus-normal estimate was +0.0262, or 2.62 percentage points. This is close in magnitude to the +3.13-point Study 1 estimate. However, Study 2 included only five measures, so its 95% interval was wider: [-0.0115, 0.0639]. Three measures favored critic debate and two favored normal debate. Importantly, leave-one-measure-out estimates stayed positive after removing any one measure, ranging from +0.0133 to +0.0413. Component-weighted preservation again showed a similar positive critic effect (+0.0315, 95% CI [-0.0106, 0.0737]).

Claude also showed a much larger benefit from independent aggregation than GPT. Independent ensemble minus single agent was +0.0999, with a 95% interval of [0.0441, 0.1695], and all five measures favored the ensemble. Normal debate was again lower than independent aggregation on average (-0.0150), but its interval [-0.0622, 0.0244] included zero.

4.5 A preservation–distortion trade-off in Claude

The most important difference between the two studies appeared in distortion. In Claude, critic debate increased mean distortion severity by 0.0362 relative to normal debate, with a 95% interval of [0.0148, 0.0576]. The any-distortion rate increased by 0.0326 [0.0104, 0.0536]. The material-distortion difference was smaller, +0.0036, with an interval that reached zero [0.0000, 0.0076].

In other words, the Claude critic condition tended to keep more information, but it also introduced more distortion at the same time. This is why preservation and accuracy should not be collapsed into one concept. A system that mentions more relevant details is not necessarily safer or more reliable if some of those details are changed in the process and represented incorrectly.

4.6 Cross-model summary

Outcome	Study 1: GPT-5.6 Sol	Study 2: Claude Sonnet 5	What I take from it
Critic – normal, claim-weighted	+0.0313 [0.0053, 0.0634]	+0.0262 [−0.0115, 0.0639]	Same direction and very similar point estimate; Claude is less precise with five measures.
Independent – single	+0.0126 [−0.0002, 0.0260]	+0.0999 [0.0441, 0.1695]	Positive in both, but much larger for Claude.
Normal – independent	−0.0027 [−0.0096, 0.0040]	−0.0150 [−0.0622, 0.0244]	Same negative direction; neither study shows a reliable advantage for ordinary debate.
Critic – normal, component-weighted	+0.0352 [0.0097, 0.0678]	+0.0315 [−0.0106, 0.0737]	Again, similar positive magnitude.

Table 3. Cross-model comparison of the planned preservation contrasts.

5. Discussion

5.1 What the results mean

The results point to a more specific conclusion than 'multi-agent systems are better' or 'debate causes groupthink.' The strongest evidence is for role structure. In Study 1, ordinary debate did not reliably outperform independent aggregation, but introducing a source-grounded critic did improve preservation. The smaller Claude replication produced almost the same critic effect magnitude, but also with wider uncertainty.

This suggests that interaction by itself may not be the valuable aspect. Independent agents can already contribute complementary information, but once agents begin revising in response to one another, that information may be compressed, repeated, or omitted. A critic changes the incentives of one participant, instead of trying to produce another generally good answer, it is explicitly asked to search for what the group is losing or stating without enough support. The condition comparison shows that the role change mattered for final-answer preservation in Study 1 and had a similar estimated effect in Claude.

5.2 Why the Claude replication helps and why it does not “prove” the result

The replication makes the paper stronger because it reduces the chance that the central pattern was unique to one model family. The critic-minus-normal point estimates were +0.0313 for GPT and +0.0262 for Claude, a difference of only about half a percentage point. The component-weighted estimates were similarly close. That cross-model similarity is encouraging.

At the same time, the Claude study was intentionally small: only five measures. Its confidence interval included zero, and individual measure effects were heterogeneous. I therefore do not want to describe it as a second confirmatory proof. I think it is better understood as directional and magnitude-level replication evidence. The right next step is a larger cross-model study.

5.3 The critic may have a cost

The Claude distortion result prevents a simplistic conclusion that the critic condition is simply best. In Claude, critic debate preserved more information but also produced more distorted components. One possible explanation is that a

critic pushes the system to include additional details and edge cases, creating more opportunities for imperfect restatement. Another is that the critic role changes the synthesis in a model-specific way. The current data cannot distinguish these explanations.

Practically, this means a future multi-agent system should not optimize only for coverage. A critic can be useful, but it should ideally be paired with a separate grounding or verification step. The more general lesson is that interventions should be evaluated on several dimensions at once: what they preserve, what they distort, and what they add without support.

5.4 Limitations

This study has several important limitations, and they shape how strongly the results should be interpreted.

- Domain scope. CA-TradeoffEval uses California ballot-measure materials. Policy documents are useful because they contain many trade-offs, but results may not generalize to science, law, medicine, or open-ended research tasks.
- Model scope. Study 1 uses one GPT model family and Study 2 uses Claude Sonnet 5 on only five measures. More models and larger replications are needed before claiming a general multi-agent effect.
- Evaluation construction. I created most of the reference inventories myself. Although the final scoring was blinded, independent experts have not yet fully audited whether the inventories capture all important source information or whether every atomic scoring rule is ideal.
- Human scoring. Final labels were human-verified, but the main confirmatory dataset was scored by one researcher with model suggestions visible. The M05 pilot showed perfect model-human agreement on 128 mixed positive/negative labels, but a second independent human scorer would provide stronger reliability evidence.
- No independent-plus-critic control. The experiment compares critic debate with normal debate, but it does not include a condition where an independent ensemble receives the same critic audit without ordinary interaction. Therefore, I cannot tell whether the critic specifically repairs interaction-induced loss or simply improves any synthesis by doing an extra source check.
- Mechanism analysis deferred. I saved intermediate outputs, but this paper does not yet quantify discovery, loss, critic recovery, and synthesis uptake at each stage. The final-output comparison therefore establishes a protocol effect more clearly than a mechanism.
- Small number of inferential units. Study 1 has ten policy measures and Study 2 has five. The thousands of component rows improve measurement resolution, but they are not thousands of independent experiments. Uncertainty should therefore remain calibrated to the number of measures.

5.5 Future work

- Independent annotation audit: ask additional annotators to build or audit inventories from the official sources without seeing condition results.
- Second human scorer: independently score a stratified subset and report raw agreement and Cohen’s kappa.
- Full cross-model replication: extend Claude to all ten measures and add additional model families.
- Mechanism analysis: use the saved intermediate outputs to measure discovery, propagation, loss, critic recovery, and whether recovered information reaches the final synthesis.
- Additional control: add an independent-ensemble-plus-critic condition to separate a general audit benefit from an interaction-specific repair effect.
- Broader tasks: repeat the study on scientific evidence synthesis, regulatory text, and other source-grounded long-form tasks.
- Joint quality metrics: add unsupported-claim, citation-grounding, length, and cost outcomes so preservation gains can be evaluated against accuracy and resource trade-offs.

5.6 Practical implications and theory of change

The immediate design lesson is modest, developers should not assume that more discussion automatically beats keeping independent analyses separate. A practical system could preserve independent drafts, give a dedicated critic access to the source, and ask the final synthesizer to account explicitly for critic-identified omissions. The Claude

results suggest that such a design should also include a verification step to catch distortions introduced while expanding coverage.

If future work validates these findings across annotators, models, and domains, CA-TradeoffEval could become one diagnostic tool for comparing multi-agent protocols. The larger goal is not to optimize one benchmark score. It is to make information loss visible enough that designers can notice and fix it before multi-agent systems are used in settings where missing a qualification or trade-off could meaningfully change a person's decision.

6. Conclusion

This project asked a simple question, when several AI agents work together, does useful information survive the conversation? Across ten California policy measures, I found that combining independent analyses produced a small positive preservation estimate over one agent, while ordinary debate did not reliably improve on that independent baseline. The clearest Study 1 result came from structured criticism: replacing one ordinary revision role with a source-grounded critic improved claim-weighted preservation by 3.13 percentage points, and the effect remained positive under alternative weighting and every leave-one-measure-out check.

A separate five-measure Claude Sonnet 5 replication produced a similar +2.62-point critic effect, although its wider interval included zero. It also exposed a possible cost: critic debate increased distortion. Together, the studies suggest that the architecture of multi-agent reasoning matters more than the fact that multiple agents are present. Independent viewpoints can help, ordinary interaction is not automatically beneficial, and explicit dissent can preserve information, but only if preservation is evaluated along with accuracy.

The main contribution is a controlled way to see an otherwise easy-to-miss failure mode in multi-agent systems, and evidence that a small change in role structure can meaningfully change what information reaches the final answer.

References

- California Secretary of State. (n.d.). Official Voter Information Guides and archived statewide election materials. California Secretary of State.
- Du, Y., Li, S., Torralba, A., Tenenbaum, J. B., & Mordatch, I. (2024). Improving factuality and reasoning in language models through multiagent debate. *Proceedings of the 41st International Conference on Machine Learning*, PMLR 235, 11733–11763.
- Irving, G., Christiano, P., & Amodei, D. (2018). AI safety via debate. *arXiv:1805.00899*.
- Liang, P., Bommasani, R., Lee, T., et al. (2022). Holistic evaluation of language models. *arXiv:2211.09110*.
- Lorenz, J., Rauhut, H., Schweitzer, F., & Helbing, D. (2011). How social influence can undermine the wisdom of crowd effect. *Proceedings of the National Academy of Sciences*, 108(22), 9020–9025. <https://doi.org/10.1073/pnas.1008636108>
- Min, S., Krishna, K., Lyu, X., Lewis, M., Yih, W.-t., Koh, P., Iyyer, M., Zettlemoyer, L., & Hajishirzi, H. (2023). FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. *Proceedings of EMNLP 2023*, 12076–12100. <https://doi.org/10.18653/v1/2023.emnlp-main.741>
- Raji, I. D., Bender, E. M., Paullada, A., Denton, E., & Hanna, A. (2021). AI and the Everything in the Whole Wide World Benchmark. *NeurIPS Datasets and Benchmarks Track*. *arXiv:2111.15366*.
- Wang, X., Wei, J., Schuurmans, D., Le, Q. V., Chi, E. H., Narang, S., Chowdhery, A., & Zhou, D. (2023). Self-consistency improves chain of thought reasoning in language models. *International Conference on Learning Representations*.
- Weng, Z., Chen, G., & Wang, W. (2025). Do as we do, not as you think: The conformity of large language models. *ICLR 2025*. *arXiv:2501.13381*.
- Wynn, A., Satija, H., & Hadfield, G. K. (2025). Talk isn't always cheap: Understanding failure modes in multi-agent debate. *arXiv:2509.05396*.
- Yin, Z., Sun, Q., Chang, C., Guo, Q., Dai, J., Huang, X., & Qiu, X. (2023). Exchange-of-Thought: Enhancing large language model capabilities through cross-model communication. *Proceedings of EMNLP 2023*, 15135–15153.
- Zhang, H., Cui, Z., Wang, X., Zhang, Q., Wang, Z., Wu, D., & Hu, S. (2025). If multi-agent debate is the answer, what is the question? *arXiv:2502.08788*.
- Zhang, J., Xu, X., Zhang, N., Liu, R., Hooi, B., & Deng, S. (2024). Exploring collaboration mechanisms for LLM agents: A social psychology view. *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 14544–14607. <https://doi.org/10.18653/v1/2024.acl-long.782>

Declarations

Data and Code Availability

The study was designed with a reproducibility and audit trail, including frozen protocols, evaluation inventories, scoring files, model outputs, and analysis artifacts. Materials that can be shared publicly will be archived in a public repository associated with this preprint. A repository URL and persistent identifier can be added to a later version after deposit.

Funding and Acknowledgments

This work was developed during the Non-Trivial Fellowship. The author acknowledges the feedback and support received during the fellowship.

Competing Interests

The author declares no competing interests related to this work.

Author and Affiliation

Spriha Pradeep is listed as an independent researcher for this preprint.

ORCID iD: 0009-0009-9451-7660

Appendix A. Reproducibility and audit trail

Study 1 used an explicit pre-unblind workflow. All ten measures were human-reviewed and locked before condition mappings were opened. A final pre-unblind integrity manifest covered all 5,872 scoring rows and confirmed zero unresolved flags. Because the repository had not previously fixed the aggregate distortion summaries or bootstrap seed/count, those choices were recorded in a separate pre-unblind analysis amendment before mapping access.

Study 2 followed the same principle. Its five-measure replication protocol was frozen before Claude generation. Study 2 outputs, scoring logs, and locks were kept separate from Study 1. All 3,840 scoring rows were human-reviewed and locked before the Claude condition mapping was opened. Major protocol, scoring, and analysis artifacts were hashed to make later changes detectable.

Key Study 1 analysis choices: four paired trials per measure-condition; measure-level inference; 10,000 paired bootstrap replicates; seed 20260809; 95% percentile intervals; leave-one-measure-out sensitivity.

Key Study 2 choices: five deterministically selected measures; same four-condition protocol; Claude Sonnet 5 generation; same frozen component inventories and scoring prompt; same bootstrap implementation reused for comparability.

Appendix B. Interpretation boundaries

Finding	What the data support	What they do not support
Independent aggregation	A positive preservation estimate in both model families; clearly positive in Claude and uncertain in Study 1.	A universal claim that ensembles always outperform single agents.
Normal debate	No reliable preservation advantage over independent aggregation; point estimates were negative in both studies.	A claim that ordinary debate systematically causes information loss.
Source-grounded critic	The strongest Study 1 preservation gain and a similar positive point estimate in Claude.	A universal critic benefit or proof of a human-like dissent mechanism.
Distortion	Little clear trade-off in Study 1; higher distortion under critic debate in Claude.	A claim that higher preservation is automatically safer or more accurate.

Table A1. Claims the current evidence can and cannot support.